

Algebra II with Trigonometry

Chapter 6.4

Olympiad Problems (GPT-5.4)

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Problems

1. Find all real numbers x in $[0, 2\pi)$ such that

$$\tan x + \cot x = 2 \sec 2x.$$

2. Let

$$f(x) = \sec x + \csc x.$$

Find the minimum value of $f(x)$ over all real x for which it is defined.

3. How many vertical asymptotes does the graph of

$$y = \tan(2x) + \cot(3x)$$

have on the interval $(0, \pi)$?

4. Determine all real x satisfying

$$\sec x - \tan x = \csc x - \cot x.$$

5. A function of the form

$$y = a \tan(k(x - b))$$

has consecutive vertical asymptotes at $x = \frac{\pi}{12}$ and $x = \frac{5\pi}{12}$, and its graph passes through $(\frac{\pi}{4}, -3)$. Find a , k , and one possible value of b .

6. Find the exact value of

$$\sec \frac{\pi}{12} \csc \frac{5\pi}{12} + \sec \frac{5\pi}{12} \csc \frac{\pi}{12}.$$